

PROFESSIONAL SUMMARY

Staff Software Engineer with **9+ years of total software engineering experience** (including 5.5+ years at Visa Inc.) architecting and delivering cloud-native platforms for real-time payment fraud detection at global scale. Extensive experience designing cloud data ingestion REST APIs, building Kubernetes infrastructure (Helm charts, ENV0/Terraform IaC), engineering GenAI-powered tooling, and owning end-to-end system architecture across AWS. Contributor to a **USPTO-granted patent** (Release Management Automation). Proficient Gen AI practitioner: daily user of **Claude, Cursor, and Cline** for full-stack development, infrastructure scripting, architecture design reviews, and technical documentation. Core technical strength in **Java 17, Python, Go, Kafka, AWS (15+ services)**, and high-throughput distributed systems. Certified by **Caltech, UT Austin, Stanford, and AWS**.

TECHNICAL SKILLS

Languages: Java 17, Python, Go, Scala, JavaScript, TypeScript

Gen AI & ML: Claude, Cursor, Cline, MCP Servers, AI-assisted development (Go, Python, shell, Helm), architecture design reviews, technical documentation, AIP REST inference, OPA/Rego, EMR retraining

Cloud (AWS): EC2 (Nitro Enclaves), EKS, Lambda, EMR, MSK (Kafka), ElastiCache, AuroraDB, S3, API Gateway, KMS, CloudWatch, CodeBuild/Pipeline

Testing Frameworks: JMeter, Blazemeter, Selenium, REST Assured, Cypress, TestNG, Cucumber, JUnit, Postman

DevOps & Infra: Kubernetes (EKS), Helm, Terraform / ENV0 (IaC), Jenkins, CodeBuild/Pipeline, Artifactory, Git (GitHub Enterprise)

Databases: MySQL, PostgreSQL, Cassandra, Redis, Valkey, Neo4j, MongoDB, HDFS / Delta Lake

Performance Testing: Blazemeter (built from scratch), JMeter, RedButton (Active-Active failover)

Security: mTLS, HMAC, AES-GCM (VisaKP), PGP, AWS KMS, Nitro Enclaves, SSL/TLS, OPA/Rego, Vault

Monitoring: CloudWatch, Grafana, Prometheus, Splunk, Humio

Messaging & Streaming: Apache Kafka (MSK), MirrorMaker2, Kinesis Firehose, SNS, Spring Boot, Apache Spark

Methodologies: Agile/Scrum, Shift-Left Testing, CI/CD, Active-Active Architecture, ITDR

PROFESSIONAL EXPERIENCE

Staff Software Engineer | Visa Inc. | Austin, TX (5 yrs 9 mos)

November 2020 – Present

Promoted progressively: Staff Software Engineer (Aug 2024 – Present) | Staff SDET (Apr 2023 – Aug 2024) | Sr. SDET (Nov 2020 – Apr 2023).

Engineered cloud-native platform services, REST APIs, IaC automation, and led technical delivery.

Cloud-Native Fraud Scoring Platform — AWS (Global Payments)

Aug 2024 – Present

Engineering lead for a dual-region Active-Active real-time fraud scoring system handling UK, Brazil, Argentina, and Costa Rica payment corridors across three AWS regions — Ireland, London, and São Paulo.

Cloud Engineering & API Design

- Engineered the VADP Decision Service (Go, EKS, gRPC), scaling the dual-region Active-Active platform to process **2,500+ transactions per second (TPS)** while maintaining a strict **99.95% latency SLA of <125ms**
- Secured **70% of Argentina's national clearinghouse transaction volume** by deploying real-time ML fraud scoring APIs, successfully intercepting millions of dollars in potential fraudulent transactions
- Architected the Cloud Data Ingestion REST API (authored Component Operations Map), replacing legacy SFTP batch processing to completely eliminate 24-hour data lag for clients in Costa Rica and New Zealand
- Seamlessly validated the **ElastiCache to Valkey migration** without downtime, optimizing high-speed in-memory data retrieval for **200M+ daily caching events** to feed the ML decision engine
- Built **Config Stack IaC environment from scratch**: authored Helm charts, configured AWS Secrets, and executed Terraform/ENV0 in precise dependency sequences (EKS → Valkey → MSK → CodeBuild)
- Identified and resolved **Lambda Authorizer cold-start latency spikes** (7–100ms+) by driving an architectural shift to move HMAC authentication into AWS Nitro Enclaves, permanently reducing API Gateway latency
- Engineered robust real-time ingestion pipelines across **5 MSK Kafka topics**, enforcing AES-GCM encryption, KMS key rotation, and zero PII leakage across all CloudWatch log streams

Gen AI Engineering

- ⚡ Daily user of **Cursor and Cline** for 2+ years: accelerated **full-stack development** (Go, Java, Python, shell, Scala) through AI-assisted script authoring, Helm/Terraform/ENV0 module generation, and rapid navigation of 20+ service repositories to conduct root cause analysis.
- ⚡ Used **Claude** extensively (1+ year) as a core **development and design tool**: conducting **architecture and API design reviews** (REST contracts, data pipeline schemas, system component design); generating **production-ready** Go/Python service code and Kubernetes configurations; and authoring technical documentation (runbooks, architecture diagrams) or test plans for complex cloud scenarios.
- ⚡ Introduced **Claude Code** to A2A Champions team (TLRTPCC-2547): configured **MCP server integrations** (Jira, Jenkins, Wiki) to enable AI-driven development workflows and pipeline automation; built a **GenAI performance results analyzer** (Go/Python) that parses JMeter/Blazemeter output to generate actionable latency insights.

Performance Engineering & Infrastructure

- Built **Blazemeter performance testing infrastructure from zero** for all environments and AWS regions; established SLAs (500 TPS sustained, 1,000 burst, P99.95 ≤ 250ms); executed 8+ campaigns — **6M+ transactions/session, zero errors, P99.95 = 104.96ms**

- Identified **Lambda Authorizer cold-start latency spikes** (7–100ms+) as production regression root cause; drove architectural change moving HMAC auth into Nitro Enclave, permanently reducing API Gateway latency
- Validated A2A Gateway (Go, Nitro Enclave), VADP Decision Service (Go, EKS, gRPC), MSK Kafka pipelines, ElastiCache → Valkey migration, ITDR readiness (MirrorMaker2, cross-region failover, GTM), and zero PII leakage across all CloudWatch log streams
- Validated all five Kafka data ingestion pipelines (TransactionStatus, Fraud, AccountDetails, Entitlements, HistoricalData) — V1/V2 detection, AES-GCM PII encryption, account concatenation, KMS key management, and PAN-Data 3-day flow (SFTP → PGP decrypt → Nitro Enclave)
- Mentored **full QA team** (~1 year): Blazemeter, CloudWatch/Grafana, Active-Active test strategy, EMR Lambda validation, release sign-off; led formal KT sessions ensuring team independence

Stack: Java 17, Python, Go, Scala, AWS (EKS, EC2 Nitro Enclaves, MSK, Lambda, EMR, ElastiCache/Valkey, AuroraDB, S3, API Gateway, KMS, CloudWatch, CodeBuild), Kubernetes, Helm, Terraform/ENV0, OPA/Rego, Blazemeter, JMeter, Redis, gRPC, Spring Boot, Apache Kafka, MirrorMaker2, Kinesis Firehose, Claude, Cursor, Cline

On-Premise Real-Time Payment Risk Platform

Apr 2023 – Aug 2024

Staff SDET on Visa's on-premise real-time fraud scoring system — sub-250ms latency, Active-Active dual datacenter architecture supporting UK and Argentina payment corridors.

- Engineered Java 17 and Python service components; executed **sustained 1,600 TPS performance tests** over 3 hours — **16.9M transactions, P99.95 = 36ms, zero errors**; ran 12+ performance campaigns at 300–1,600 TPS with zero 5xx errors
- Validated end-to-end **VisaKP cryptographic migration** (legacy encryption → one-way HMAC) across Java ALR Consumer (in-place Redis re-encryption), Fraud Consumer (MySQL rebuild), and online Decision Service DAG; confirmed HMAC key determinism for correct GFL/ALR scoring
- Led **SSL/TLS weak cipher remediation** across 3 Java/REST APIs; validated Cassandra TLS upgrade (zero downtime), Java SLP security findings (2 batches), Oracle MySQL Critical Patches (CPUJAN2024, CPUJUL2024)
- Validated **43 Risk Indicators (rskInd)** in scoring API — Consul mock mode and live AIP integration — with schema propagation through RaaSOrchestrator, Hadoop consumers, billing, and reporting pipelines
- Led AIP transaction count mismatch investigation across VDP, RaaS HDFS, and AIP; identified timing and retry-logic root causes; unblocked production release
- Validated Global Risk Accounts, GFL Rate Limit, AIP Scoring Consumer scalability redesign (Spark 3.4→3.5, 4-processor architecture), V1/V2 API changes, request deduplication, and **Java 17 migration**

Stack: Java 17, Python, Go, Scala, Apache Spark 3.4/3.5, Delta Lake, HDFS, Apache Kafka (VRM), Redis, Cassandra, MySQL, Consul, Vault, Humio, Splunk, Grafana, Prometheus, JMeter, Jenkins, Artifactory, VisaKP/HMAC, SSL/TLS

Enterprise Fraud Management — Decision Manager UI

Nov 2020 – Apr 2023

Sr. Software Development Engineer in Test across Merchant & Acquiring Processing and Visa Risk departments.

- **Built FME Dashboard automation from scratch using Java and Python:** full suite covering table rendering, pagination, filters, and UI-API validation; adopted by wider QA team
- Found **critical Processor Config API (PCFGAPI) data integrity bug:** PATCH calls silently accepted invalid processor attributes (e.g., invalid fields for Citicon, Barclays), risking production merchant configuration corruption; validated fix pre-production
- Identified and validated fixes for **30+ UI bugs:** browser compatibility (IE/Safari/Firefox), timezone mismatch in IMS Flag calculations, WCAG 2.1 accessibility failures (400% reflow), and localization issues across 5+ report types

Stack: Java, Python, Angular, React, JavaScript, TypeScript, Selenium WebDriver, WebdriverIO, REST Assured, Postman, Jenkins, JIRA, HTML5, CSS3

Software Development Engineer in Test | FINRA | Rockville, MD

July 2019 – January 2021

- Led testing for financial surveillance platforms (including Behaviour Analytics and Regulatory Data pipelines); implemented Page Object Model automation using Selenium WebDriver, Java, Selenium Grid, Cucumber, TestNG, and Maven
- Executed parallel cross-browser testing and automated test execution via Linux shell scripts; evaluated CI/CD pipeline severity for build releases

Stack: Java, Python, MySQL, PostgreSQL, REST Assured, TestNG, Selenium WebDriver, Selenium Grid, Cucumber, Jenkins, Apache JMeter, Maven, Linux

Full Stack Developer | ADP | Hyderabad, India

August 2016 – July 2017

- Developed chat functionality for IPM HR services platform using ReactJS, Node.js, and RESTful APIs; reduced page load time to under 2s for 10+ websites, boosting conversion by up to 37%
- Built Selenium automated test scripts with cross-browser parallel testing via Selenium Grid and JUnit

Stack: Java, Python, ReactJS, Node.js, JavaScript, MongoDB, RESTful APIs, Selenium WebDriver, Selenium Grid, JUnit, HTML5, CSS3, Webpack

PATENT

Release Management Automation — "Release Management as a Service"

USPTO — Granted | Visa Inc.

Co-inventor: automated pipeline from RISKRM Jira ticket state transition → JSON deployment config → Jenkins CI/CD for automated NonProd deployments, BVT execution, and Jira status tracking. Reduced deployment time by **20%**, eliminated manual steps, established single-source-of-truth release management with automatic audit trail.

EDUCATION & CERTIFICATIONS

Post Graduate Program in AI & Machine Learning | McCombs School of Business, UT Austin

Dec 2025 – Aug 2026

AI & Machine Learning Bootcamp | California Institute of Technology (Caltech)

Dec 2024 – Nov 2025

AWS Certified Developer – Associate | Amazon Web Services

July 2021

Machine Learning Specialization | Stanford University (Online)

Feb 2023

Master of Science, Computer Science | George Mason University | Fairfax, VA

May 2019

Bachelor of Technology, Computer Science | Mahatma Gandhi Institute of Technology | Hyderabad, India

May 2017